LAHOUCINE AZMOUD

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Summary

Backend Software Engineer and **1337 (42 Network)** student with hands-on experience building **secure, high-performance APIs, automation daemons, and network-level services** from scratch. Passionate about **security engineering** — from designing hardened authentication flows and scanning-pipeline tooling to low-level network programming and event-driven architectures. Proficient in **Python, C/C++, Rust, Go, TypeScript, and Bash**, with a strong habit of **AI-assisted development**, rigorous peer code review, and automated testing. Ready to contribute to production security systems from day one.

Education

1337 Coding School — 42 Network *Casablanca, Morocco* *2024 – Present*
Computer Science & Software Engineering

- Specialized in **Linux internals, socket/network programming, low-level memory management**, concurrent architectures, and secure systems design.
- Engaged in intensive **peer code reviews**, algorithm design, software testing, and collaborative engineering pipelines — mirroring professional team workflows.

Ibn Al Haytham High School *Physics* *2017 – 2020*

Technical Projects

webserv *High-Performance HTTP/1.1 Server from Scratch* *C++, Unix, epoll*

- Built a fully **non-blocking, event-driven HTTP server** in C++ using `epoll` I/O multiplexing and raw TCP sockets — gaining deep understanding of **network-layer request parsing**, connection lifecycle, and attack-surface exposure.
- Implemented complete **HTTP/1.1 protocol parsing**, request routing, dynamic **CGI execution pipelines**, and resilient memory management — forming a practical foundation for **static and dynamic application analysis**.

clippy_hook *X11 Clipboard Automation Daemon* *Rust, Python, Linux/X11, Unix Sockets*

- Developed a **high-performance Rust daemon** that continuously monitors X11 clipboard streams, detects copied media URLs, and dispatches background automation scripts via **Unix domain sockets** — architected as a **scanning pipeline** with decoupled detection and execution stages.

skilio *Full-Stack Freelance Marketplace Platform* *Python, FastAPI, React 19, TypeScript, PostgreSQL, Redis, Docker*

- Engineered **multi-provider OAuth 2.0 (Google & LinkedIn) and JWT authentication** with CSRF state validation, Argon2 password hashing, and role-based access control — designing **secure API endpoints** resistant to common authentication attacks.
- Built a **real-time WebSocket messaging system** backed by a **Redis Pub/Sub event bus**, with a comprehensive `pytest` + **GitHub Actions CI** test suite covering authentication, chat, and all API endpoints.
- Integrated **Cloudinary CDN endpoints** for secure media upload/download

Blade *TUI Text Editor with Custom Tokenizer* *C, ncurses*

- Built a terminal text editor in C with a **hand-written tokenizer and lexer**, modal editing, an FFT music visualizer, and an embedded file browser — demonstrating sustained low-level systems engineering in a production-quality C codebase.

Professional Experience

Hztech *Full Stack Developer Intern* Casablanca, Morocco *May 2022 – Sep 2022*

- Engineered and shipped the company's web application using **Golang, Svelte, and SQLite**, optimizing database queries and contributing directly to a **production system**.
- Built reliable UI components for an electronic software management solution with clean API integration and participated in **active code reviews and continuous debugging** in a collaborative startup environment.

Skills & Competencies

Languages: Python **(primary)**, C/C++, Rust, Go, TypeScript, JavaScript, SQL, Bash
Security & Systems: OAuth 2.0 / JWT / CSRF, API Security Hardening, Linux/Unix Internals, Socket & Network Programming, Static Code Tokenization, Process Isolation, systemd
Backend & Infrastructure: FastAPI, PostgreSQL, Redis, Docker, REST APIs, WebSocket / Event-Driven Architecture, CI/CD (GitHub Actions)
Engineering Practices: AI-Assisted Development & Code Review, Automated Testing (pytest), Peer Code Review (42 Network), Data Structures & Algorithms, Debugging & Profiling
Spoken Languages: English *(Fluent/Professional)*, French *(Intermediate)*, Arabic *(Native)*, Tamazight *(Native)*